Current: charge flow rate

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Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate current as charge transferred per time.

SUCCESS CRITERIA - I can...

1. Use I=q/t with seconds and coulombs.
2. Describe conventional current direction.
3. Explain steady charge conservation at a junction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “current” and “charge” to explain your answer. Sentence starters are optional.

current: rate of charge flow

charge: a property responsible for electric interactions

interval: the time or range between two stated values

01 Warm-up: bring back an earlier idea

Both charge magnitudes become **3** times as large and separation becomes **5** times as large. Express new force as a fraction of the old.

Word help: separation = distance between the specified objects or centers

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: I = Δq/Δt.
Current is charge per time; **1 A** = **1 C/s**. Conventional current follows positive charge motion.

Worked reasoning: If **12 C** pass a point in **4 s**, average current is I=Q/t=**3 A**. Charge circulates; a resistor does not consume charge.

Your turn: Draw a complete loop and **label equal current before** and after one resistor in steady state.

Sentence starter: In my work, I used … to show … . I labeled it … .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Draw conventional current around the** closed circuit, outside the battery from + toward -.
2. **What happens** if a wire is disconnected?

Sentence starter: In my work, I used … to show … . I labeled it … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

Current: charge flow rate

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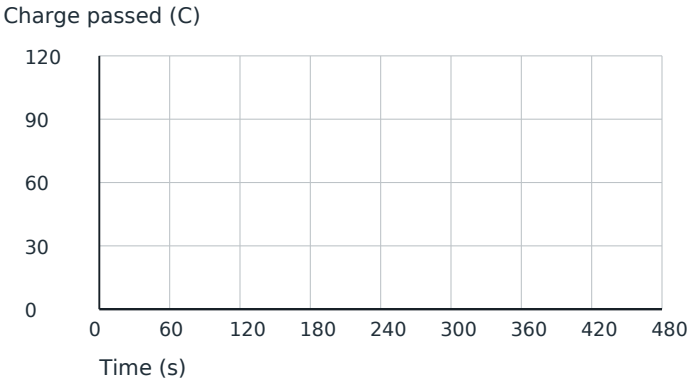
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A wire carries a steady **0.25 A** for **8 minutes**. Convert time to seconds, then **calculate charge transferred**. **Draw a cumulative-charge graph** and **label the slope**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

At a junction, **5 A** enters and **2 A** leaves along one branch.
1. **Find the other outgoing current** in steady operation.
2. **Explain the conservation rule**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A source approaching a stationary listener at **20 m/s** emits **600 Hz**. Use v_{sound}=**340** and $f_{obs}=f_s\ v/(v-v_s)$. **Find observed frequency**.
Use wavefront spacing to **explain the frequency change**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Source moves right. Add wavefronts.



Marzano check - circle: ① ② ③

07 Exit check

Does a large electron drift speed have to equal the speed at which a switched circuit responds? **Explain qualitatively**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



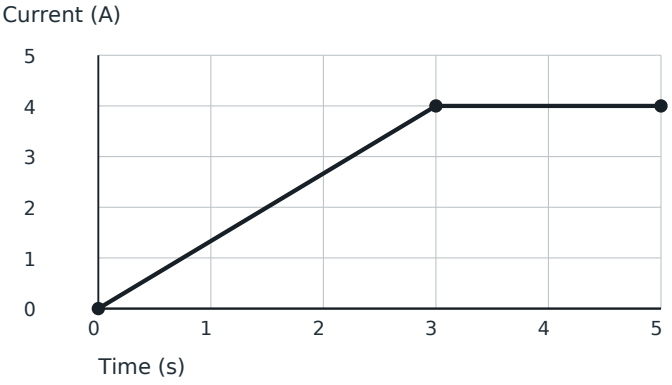
Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A current rises linearly from **0** to **4 A** during **3 s**, then stays **4 A** for **2 s**. **Find total charge** from the current-time graph.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③